

AI CEO: Strategic Intelligence Agent

Duration

1 Week

Type

Individual Project

Background

Organizations today operate in an environment characterized by continuous information flow. News articles, analyst reports, company announcements, market developments, customer opinions, technological innovations, and competitor activities create an overwhelming amount of information.

The challenge is no longer finding information.

The challenge is transforming information into strategic decisions.

In this project, you will build an AI-powered Strategic Intelligence Agent that continuously collects, analyzes, and reasons over live information related to a selected company and provides executive-level recommendations.

Your system should behave as an intelligent advisor to the CEO.

The objective is not to summarize information, but to generate actionable business insights supported by evidence.

Scenario

You have been hired as an AI Strategy Consultant by the executive board of a company.

Your task is to develop an AI-powered Strategic Intelligence Agent capable of:

- Collecting live information from multiple public sources
- Monitoring the company's environment
- Identifying opportunities and threats
- Detecting emerging trends
- Generating strategic recommendations
- Explaining decisions using supporting evidence

At the end of the project, your system should be able to answer:

“If you were the CEO today, what would you do next and why?”

Step 1: Select a Company

Choose one publicly visible company.

Examples:

- Tesla
- NVIDIA
- SAP
- DHL
- BMW
- Lufthansa
- Siemens
- Airbus

You may propose another company upon approval.

Project Objective

Develop an AI Strategic Intelligence Agent that automatically gathers live information and transforms it into executive-level recommendations.

The system should answer questions such as:

- What are the major opportunities for the company?
- What are the biggest risks?
- What are competitors doing?
- Which technologies or trends should management monitor?
- What strategic actions should be prioritized?
- What evidence supports these recommendations?

Functional Requirements

Task 1: Live Data Collection

The system must automatically collect information from at least **three public sources**.

Possible sources include:

Company Sources

- Press releases
- Investor relations pages
- Official announcements

News Sources

- Financial news
- Industry news
- Technology news

Market Sources

- Competitor websites
- Product announcements
- Industry reports

Community Sources

- Reddit discussions
- Product reviews
- Public forums

Research Sources

- Research publications
- Technical blogs
- Innovation reports

Minimum Requirement

- At least 100 collected documents/articles/posts
- At least 3 independent information sources
- Automatic collection process

Task 2: Knowledge Repository

The collected information must be stored and indexed.

Possible approaches:

- FAISS
- ChromaDB
- PostgreSQL
- SQLite
- Knowledge Graph

The repository should support efficient retrieval during analysis.

Task 3: Information Processing

Your system should automatically:

- Clean data
- Remove duplicates
- Extract relevant information
- Generate embeddings
- Index information for retrieval

Task 4: Strategic Intelligence Engine

The system should identify:

Opportunities

Examples:

- Emerging technologies
- New markets
- Partnerships
- Product opportunities

Risks

Examples:

- Competitive threats
- Regulatory changes
- Negative public sentiment
- Supply chain issues

Trends

Examples:

- Technology trends
- Customer behaviour shifts
- Industry developments

Task 5: AI CEO Agent

The AI CEO Agent should:

- Analyze collected information
- Reason about business implications
- Prioritize actions
- Recommend strategic decisions
- Justify recommendations

The goal is not information retrieval.

The goal is strategic decision-making.

Task 6: Evidence-Based Recommendations

Every recommendation must contain:

Recommendation

Example:

Increase investment in AI-powered products.

Supporting Evidence

- Evidence Source 1
- Evidence Source 2
- Evidence Source 3

Expected Impact

- Revenue growth
- Market differentiation
- Customer acquisition

Risk Assessment

- Financial risk
- Operational risk
- Strategic risk

Dashboard Requirements

Develop an Executive Intelligence Dashboard.

The dashboard should contain the following sections.

Section 1: Company Overview

Display:

- Company name
- Industry
- Number of collected documents
- Number of data sources
- Last update timestamp

Section 2: Market Intelligence

Display:

- Recent news
- Competitor activities
- Emerging technologies
- Important company announcements

Section 3: Opportunity Monitor

Display:

- Opportunity title
- Impact level
- Evidence
- Confidence score

Section 4: Risk Monitor

Display:

- Risk title
- Risk category
- Severity level
- Evidence

- Confidence score

Section 5: Sentiment Analysis

Display:

- News sentiment
- Public sentiment
- Sentiment trends

Include appropriate visualizations.

Section 6: Strategic Recommendations

For each recommendation display:

- Recommendation
- Priority (High / Medium / Low)
- Supporting evidence
- Expected impact
- Risk level

Section 7: CEO Briefing

Generate an executive summary answering:

What happened?

Why does it matter?

What should management do next?

Technical Requirements

Mandatory Components

Large Language Model

You must use an open-source or freely accessible language model.

Recommended models:

- Qwen 3 8B
- Llama 3.1 8B
- Mistral 7B Instruct
- Phi-4 Mini

Embedding Model

Recommended:

- BAAI bge-base-en-v1.5
- BAAI bge-small-en-v1.5
- all-MiniLM-L6-v2

Retrieval Mechanism

- RAG
- Semantic Search
- Hybrid Search

Dashboard

Examples:

- Streamlit
- Dash

Not Allowed

The following may not be used as the primary reasoning engine:

- OpenAI API models
- Anthropic API models
- Gemini API models
- Any paid commercial LLM API

The final solution must rely on open-source or freely accessible models.

Deliverables

Deliverable 1: Working Prototype (30%)

Demonstrate a fully functional Strategic Intelligence Agent capable of:

- Collecting live information
- Storing information
- Retrieving information
- Analyzing information
- Generating recommendations

Deliverable 2: Executive Intelligence Dashboard (10%)

Demonstrate an interactive dashboard suitable for executive decision-making.

The dashboard should clearly present:

- Opportunities
- Risks
- Trends
- Strategic recommendations

Deliverable 3: Architecture Documentation (10%)

README.md should include:

- System architecture diagram
- Data flow diagram
- Technology stack
- Design decisions
- AI pipeline

Final Oral Examination 4: Architecture Documentation (50%)

- Project Demonstration

- Live Coding: You will be asked to modify or extend their solution during the examination.
- Conceptual Questions to explain concepts used in their solution.